

NoteFlow — Full Stack Notes Application

Product Requirements Document (PRD)

Product Overview

NoteFlow is a modern full stack notes management application that allows users to create, organize, search, update, and manage personal notes securely.

Tech Stack

Frontend: React, React Router DOM, Axios, Tailwind CSS, Context API

Backend: Node.js, Express.js, MongoDB, Mongoose, JWT, bcryptjs

Deployment: Vercel, Render, MongoDB Atlas

Core Features

- User Authentication
 - Create, Edit, Delete Notes
 - Search and Filter Notes
 - Pin Important Notes
 - Responsive Design
 - Dark/Light Theme
 - Toast Notifications
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User Roles

Guest User: Can register and login.

Authenticated User: Can manage notes and profile.

Frontend Pages

- Landing Page
- Login Page
- Signup Page
- Dashboard
- Profile Page
- 404 Not Found Page

Database Collections

Users Collection:

name, email, password, timestamps

Notes Collection:

title, description, tags, pinned, color, userId, timestamps

API Endpoints

Auth APIs:

POST /api/auth/signup

POST /api/auth/login

GET /api/auth/profile

Notes APIs:

GET /api/notes

POST /api/notes

PUT /api/notes/:id

DELETE /api/notes/:id

Context API Structure

AuthContext: user, token, auth state

NotesContext: notes, search, filters

UIContext: theme, modal, notifications

Security Requirements

- Password hashing using bcryptjs
 - JWT Authentication
 - Protected Routes
 - Environment Variables
 - Input Validation
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Future Enhancements

- Rich Text Editor
- Markdown Support
- AI Summarization

- Collaboration Features
 - Image Uploads
 - Voice Notes
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Development Phases

1. UI Development
 2. Backend Setup
 3. Database Integration
 4. Authentication
 5. CRUD Integration
 6. Search & Filters
 7. Deployment
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Expected Learning Outcomes

- MERN Stack Architecture
 - REST API Development
 - Authentication Systems
 - MongoDB Integration
 - Context API
 - Deployment Process
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